

Joao Vitor Pires Henares

Gameplay Systems | Technical Design | Tools

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SELECTED PROJECTS

2DBonk – Award-Winning Humor-Focused Survivor Roguelike (3-Week Production, Unity)

Solo Design + Programming

- Shipped a polished prototype in **3 weeks**; selected as a **finalist** in the FIEA Industry Review and awarded **“Most Addictive Game.”**
- Designed and implemented **10 enemy types, 3 playable characters**, weapons, passives, stages, and a full XP/progression system.
- Built **custom state machines** for combat, enemy behavior, and event-driven rhythm interactions.
- Implemented **pooled enemy spawning** system supporting 1000+ concurrent enemies with stability.

RACHA – Brazilian Crime Roguelike Deckbuilder (6-Week Gauntlet-Style Production, Unity)

Solo Design + Programming

- Shipped a **Balatro-inspired roguelike** with 25+ unique agents, 10 card types, and 5 boss modifiers.
- Built a **strategy pattern architecture** where all new content is addable as new files with zero changes to existing systems.
- Designed a **typed blackboard context system** (run/level/round scoped) decoupling logic across all systems.
- Implemented a **sequential visual event queue** supporting complex multi-agent chain reactions and replay.

Ravenous Sky – 18-Person Adventure Game (FIEA Capstone, UE5)

Technical Designer, November 2025 – Present

- Contributed to the **character controller**, implementing movement logic and physics-based responsiveness.
- Designed and owned a performance-optimized **debris system**, implementing object pooling and data-driven configuration to maintain runtime stability.
- Implemented a full **dynamic soundtrack system** in Unreal using the engine’s audio framework (state-driven transitions and runtime control).
- Developed gameplay scripting and debugging tools supporting cross-disciplinary system integration.

338th Training Squadron of the U.S. Air Force – TaskForce Productivity RPG App (Unity)

Team Lead, January 2026 – May 2026

- **Led a 7-person multidisciplinary squad** from concept to delivery, owning feature planning, sprint coordination, and milestone presentations.
- **Designed and implemented core systems**: progression, faction leaderboards, and character customization.
- **Coordinated designers, engineers, artists, and production** across the full project lifecycle.
- **Built internal tooling and documentation** supporting team productivity and pipeline consistency.

Saparkour Series – Physics Traversal Prototypes (Unity)

Solo Design + Programming

Saparkour (2-week prototype)

- Built a **custom character controller** supporting grappling, bouncing, and high-speed traversal.
- Designed **5 handcrafted levels** and **15+ interactive props**.
- Implemented **physics-based grapple** mechanics and **competitive leaderboard tracking**.

Saparkour: Evolution (3-week rebuild)

- **Rebuilt the traversal system** using a fully custom Rigidbody-based controller.
- Implemented **momentum-driven** movement, **drift** steering, and spline-based level traversal.
- Designed a **dynamic grapple system** supporting **rope wrapping** and **moving grapple anchors**.

EXPERIENCE

BTG Pactual

São Paulo, Brazil

Software Engineering Intern

Summer 2023, Summer 2024

- Built an **AI-powered document-processing pipeline** that reduced manual extraction work by **90%**.
- Developed full-stack features using **TypeScript, React, and Go**, following clean architecture practices.
- Automated recurring workflows, improving team productivity and reducing manual overhead.
- Expanded experience with **AWS**, CI/CD, and production-grade engineering systems.

EDUCATION

Florida Interactive Entertainment Academy (FIEA), University of Central Florida

Orlando, FL

Master’s in Interactive Entertainment

Graduating in 2026

Focus: Gameplay systems, scripting, technical pipelines, rapid prototyping, cross-disciplinary collaboration.

University of Notre Dame

Notre Dame, IN

B.S. in Computer Science

Graduated in 2024

Focus: AI and Systems Development | GPA: 3.82 | Honors: 4 times Dean’s List nominee

TECHNICAL SKILLS & TOOLS

Engines: Unreal Engine (Blueprints + C++), Unity

Programming: C#, C++, Python, Go, TypeScript

Tech Design: Gameplay systems, tools creation, UX interaction flows, event scripting, feature documentation

Other Competencies: Multithreading, AI/ML implementation, data pipelines, networking, design patterns

Languages: Portuguese (native), English (fluent), Spanish (advanced), German (intermediate)

TECHNICAL DESIGN EXPERIENCE

Gameplay Scripting: Unreal Blueprints, Unreal C++, C#; building modular, designer-facing systems

Systems Design: Combat, movement, puzzles, AI behavior, XP/progression, procedural systems

Rapid Prototyping: Strong iteration speed – multiple prototypes per week at FIEA

Tools & Pipelines: Creating debug tools, data-driven configuration systems, designer QoL tools, Perforce

Debugging & Profiling: Performance tuning in Unreal and Unity (logs, gameplay debugger, CPU/GPU profiling)

Design Documentation: One-page design docs, feature specs, workflow documentation for teams

Cross-Disciplinary Collaboration: Translating design needs to engineering tasks; supporting designers with tools and scripting

Level/Gameplay Scripting: Triggers, interactions, event systems, scripted sequences

GAME DEVELOPMENT AND TECH DESIGN EXPERIENCE

FIEA, University of Central Florida

Orlando, Florida

Tech Designer

Since Fall 2025

- Created dozens of **rapid prototypes** exploring mechanics across genres (movement, rhythm combat, AI, physics, puzzle logic).
- Designed and implemented gameplay systems in **Unity** and **Unreal Engine (Blueprints + C++)**.
- Built internal **debugging tools**, data-driven systems, and designer-facing utilities (spawners, visualization tools, data tables, scripting helpers).
- Collaborated with multi-disciplinary teams to deliver playable builds under tight constraints.
- Authored a variety of design documentation, including **one-page feature designs, design pitch decks**, and full **Game Design Documents (GDDs)**.